

OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Parallel-Coupled Lines & Power Dividers

CGEC | B.Tech ECE (2023–27) | Biraj Sarkar

Contents

1	Parallel-Coupled Microstrip Lines	2
1.1	Even-Mode and Odd-Mode Analysis	2
1.2	Characteristic Impedances (Z_{0e}, Z_{0o}) & Coupling Equations	2
1.3	Quarter-Wave Coupling Region Length	2
2	Branch-Line Coupler and Rat-Race Hybrid Ring	3
2.1	Branch-Line Power Divider (3 dB, 90° Quadrature Hybrid)	3
2.1.1	Scattering Matrix ($[S]$) of Branch-Line Coupler	3
2.2	Rat-Race Hybrid Ring (180° Hybrid)	3

1 Parallel-Coupled Microstrip Lines

Coupled microstrip lines consist of two unshielded, parallel microstrip conductors of width W separated by a gap distance S on a dielectric substrate of thickness h . When electromagnetic signals travel on one line, energy is coupled to the adjacent line via fringing electric and magnetic fields.

1.1 Even-Mode and Odd-Mode Analysis

Any arbitrary voltage and current excitation on two symmetrical coupled lines can be decomposed into two fundamental orthogonal modes:

1. **Even Mode (+ +):** Both conductors are excited with equal amplitude and equal phase ($V_1 = V_2 = V_e, I_1 = I_2 = I_e$). No net current flows across the central plane of symmetry. The symmetry plane acts as an **ideal magnetic wall** ($\partial E / \partial n = 0, H_{tan} = 0$).
2. **Odd Mode (+ -):** Conductors are excited with equal amplitude but opposite phase ($V_1 = -V_2 = V_o, I_1 = -I_2 = I_o$). The zero-voltage plane of symmetry acts as an **ideal electric wall** ($E_{tan} = 0$).

1.2 Characteristic Impedances (Z_{0e}, Z_{0o}) & Coupling Equations

Coupled Line Impedance & Coupling Formulas

The system characteristic impedance Z_0 and voltage coupling coefficient C are uniquely determined by even-mode impedance Z_{0e} and odd-mode impedance Z_{0o} :

$$Z_0 = \sqrt{Z_{0e} \cdot Z_{0o}} \quad ; \quad C = \frac{Z_{0e} - Z_{0o}}{Z_{0e} + Z_{0o}} \quad (1)$$

Inverting these relations yields design formulas for Z_{0e} and Z_{0o} given Z_0 and coupling factor C :

$$Z_{0e} = Z_0 \sqrt{\frac{1+C}{1-C}} \quad ; \quad Z_{0o} = Z_0 \sqrt{\frac{1-C}{1+C}} \quad (2)$$

1.3 Quarter-Wave Coupling Region Length

For a single-section coupled line directional coupler, maximum power coupling occurs when the physical length L of the coupled region is a quarter guided wavelength ($\lambda_g/4$):

$$L = \frac{\lambda_g}{4} = \frac{c}{4f\sqrt{\varepsilon_{eff,avg}}} \quad \text{where } \varepsilon_{eff,avg} \approx \frac{\varepsilon_{eff,e} + \varepsilon_{eff,o}}{2} \quad (3)$$

Worked Numerical: 10 dB Coupled Line Design

Problem: Design a 10 dB microstrip directional coupler in a 50Ω system. Calculate the required even-mode (Z_{0e}) and odd-mode (Z_{0o}) characteristic impedances.

Solution:

Step 1: Convert coupling factor $C_{dB} = 10$ dB to voltage coupling ratio C :

$$\begin{aligned} C_{dB} = 20 \log_{10} \left(\frac{1}{C} \right) = 10 &\implies \log_{10}(C) = -0.5 \\ \implies C = 10^{-0.5} = 0.3162 \end{aligned} \quad (4)$$

Step 2: Calculate Z_{0e} and Z_{0o} for $Z_0 = 50 \Omega$:

$$\begin{aligned} Z_{0e} &= 50 \sqrt{\frac{1+0.3162}{1-0.3162}} = 50 \sqrt{\frac{1.3162}{0.6838}} \\ &= 50 \sqrt{1.9248} = 50 \times 1.3874 = 69.37 \Omega \end{aligned} \quad (5)$$

$$Z_{0o} = 50 \sqrt{\frac{1-0.3162}{1+0.3162}} = 50 \sqrt{0.5195} = 50 \times 0.7208 = 36.04 \Omega \quad (6)$$

Step 3: Check consistency: $Z_0 = \sqrt{Z_{0e}Z_{0o}} = \sqrt{69.37 \times 36.04} = \sqrt{2500} = 50.0 \Omega$ (Correct!).

2 Branch-Line Coupler and Rat-Race Hybrid Ring

2.1 Branch-Line Power Divider (3 dB, 90° Quadrature Hybrid)

A **Branch-Line Coupler** is a 4-port symmetrical directional coupler consisting of four $\lambda_g/4$ microstrip arms forming a rectangular ring.

Branch-Line Coupler Design Parameters

For a 3 dB equal power split in a 50 Ω system ($Z_0 = 50 \Omega$):

- Horizontal main series arms impedance: $Z_A = \frac{Z_0}{\sqrt{2}} = \frac{50}{\sqrt{2}} \approx 35.35 \Omega$.
- Vertical shunt branch arms impedance: $Z_B = Z_0 = 50 \Omega$.
- All 4 branch lengths are exactly $L = \lambda_g/4$.

2.1.1 Scattering Matrix ([S]) of Branch-Line Coupler

When signal is fed at Port 1, power splits equally between Port 2 (0°) and Port 3 (−90°), while Port 4 is fully isolated ($S_{41} = 0$):

$$[S]_{\text{Branch-Line}} = -\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix} \quad (7)$$

2.2 Rat-Race Hybrid Ring (180° Hybrid)

A **Rat-Race Coupler** consists of a circular microstrip ring of total perimeter $1.5\lambda_g$ ($6\theta = 540^\circ$) with 4 ports. Ports 1, 2, 3, and 4 are separated by $\lambda_g/4$ gaps, except between Port 3 and Port 4 which are separated by a $3\lambda_g/4$ gap.

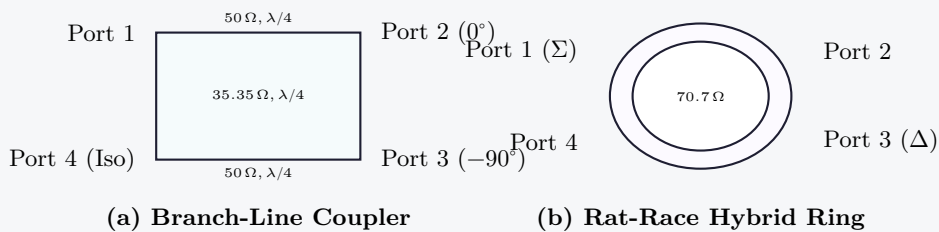
Rat-Race Impedance & Scattering Matrix

Ring characteristic impedance: $Z_{\text{ring}} = \sqrt{2}Z_0 \approx 70.7 \Omega$ (for $Z_0 = 50 \Omega$).

$$[S]_{\text{Rat-Race}} = -\frac{j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix} \quad (8)$$

- **Sum Port (Σ / Port 1):** Input at Port 1 splits equally in-phase (0°) between Ports 2 and 4. Port 3 is isolated due to destructive interference (180° path difference).
- **Difference Port (Δ / Port 3):** Input at Port 3 splits equally out-of-phase (180°) between Ports 2 and 4. Port 1 is isolated.

Topologies of Branch-Line Coupler (90°) and Rat-Race Hybrid Ring (180°)



Quick Revision — Module 4

Parallel-Coupled Microstrip Lines & Power Dividers | OE-EC804B

Key Concepts and Metrics at a Glance

System Impedance Z_0	$Z_0 = \sqrt{Z_{0e} \cdot Z_{0o}}$ (Geometric mean of even and odd mode impedances)
Voltage Coupling Factor C	$C = \frac{Z_{0e} - Z_{0o}}{Z_{0e} + Z_{0o}} \Rightarrow Z_{0e} = Z_0 \sqrt{\frac{1+C}{1-C}}, Z_{0o} = Z_0 \sqrt{\frac{1-C}{1+C}}$
Branch-Line Arm Impedances	Main series arms $Z_A = Z_0/\sqrt{2} = 35.35 \Omega$; Branch arms $Z_B = Z_0 = 50 \Omega$.
Rat-Race Ring Impedance	Ring characteristic impedance $Z_{\text{ring}} = \sqrt{2}Z_0 = 70.7 \Omega$.
Rat-Race Arm Spacings	3 sections of $\lambda_g/4$ and 1 section of $3\lambda_g/4$ (Total perimeter $1.5\lambda_g$).

One-Line Definitions

- **Even Mode (Z_{0e}):** Symmetrical mode where coupled line currents flow in identical directions with magnetic wall symmetry.
- **Odd Mode (Z_{0o}):** Anti-symmetrical mode where coupled line currents flow in opposite directions with electric wall symmetry.
- **Quadrature Hybrid (90°):** Directional coupler providing equal power division with 90° phase shift between output ports.
- **Rat-Race Ring (180°):** 4-port hybrid ring providing equal in-phase (0°) or anti-phase (180°) power division.

Frequently Asked / Viva Questions

Q1. Define even-mode and odd-mode impedances (Z_{0e}, Z_{0o}) of coupled microstrip lines.

Z_{0e} is the characteristic impedance when lines are driven in-phase with equal currents. Z_{0o} is the impedance when lines are driven out-of-phase with opposite currents.

Q2. How is characteristic impedance Z_0 related to Z_{0e} and Z_{0o} ?

$$Z_0 = \sqrt{Z_{0e} \cdot Z_{0o}}.$$

Q3. What is the physical length of a parallel-coupled microstrip directional coupler?

The coupled region length $L = \lambda_g/4$ at the center frequency.

Q4. Explain the operation of a 3 dB Branch-Line Coupler.

A 4-port quadrature hybrid consisting of four $\lambda_g/4$ branches. Input at Port 1 splits equally between Port 2 (0°) and Port 3 (-90°), with Port 4 isolated.

Q5. What are the arm impedances of a 50 Ω Branch-Line Coupler?

The horizontal series arms have impedance $Z_0/\sqrt{2} = 35.35 \Omega$, and vertical shunt branch arms have impedance $Z_0 = 50 \Omega$.

Q6. Describe the structure and port spacing of a Rat-Race Hybrid Ring.

A circular ring of characteristic impedance $\sqrt{2}Z_0 = 70.7 \Omega$ with 4 ports. Three port gaps are $\lambda_g/4$ and one gap is $3\lambda_g/4$ (total perimeter $1.5\lambda_g$).

Q7. How does the Rat-Race coupler operate as a sum (Σ) and difference (Δ) network?

Input at Port 1 splits in-phase (0°) at Ports 2 and 4 (Σ mode). Input at Port 3 splits out-of-phase (180°) at Ports 2 and 4 (Δ mode).

Frequently Asked / Viva Questions (contd.)**Q8. What causes directivity degradation in microstrip coupled lines?**

Phase velocity mismatch between even mode (β_e) and odd mode (β_o), caused by non-homogeneous substrate/air dielectric boundary.

Q9. How is crosstalk defined in parallel-coupled lines?

Crosstalk is unwanted electromagnetic power coupling from a driven primary line to an adjacent secondary signal line.

Q10. What is the bandwidth restriction of branch-line and rat-race couplers?

They are narrowband components ($\approx 10\% - 20\%$ bandwidth) because their operation relies on exact $\lambda_g/4$ line lengths.